1 Combinatorial Properties of the Configuration Structure of the 6-Dimensional Hyperrectangle

1.1 —— Spin Classification, Configuration Enumeration, and Signed Area ——

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1.2 0. Purpose of This Paper

This paper defines a 6-dimensional structure obtained by adding a discrete label axis Q to the 5-dimensional hyperrectangle, and derives the combinatorial properties arising from its configuration structure.

The main body (§1–§9) states only the mathematical structure of the model and the consequences derived purely from combinatorics. The correspondence with the Standard Model is presented in §10 as "an example of interpretation," and does not constitute a claim of this paper. It is merely an example showing how the configuration structure of the model might be mapped to the particle classification of the Standard Model.

On the external assumptions of this model: This paper takes the 5-dimensional hyperrectangle as its starting point. Why 5 dimensions, and why 3 of the 5 axes are later treated as "spatial axes," are **external assumptions of the model** and are not justified within the model itself. This dimensional setting is adopted as a correspondence to the empirical fact that the observed spacetime consists of 3 spatial dimensions + 1 temporal dimension.

1.3 1. Definition of the 6-Dimensional Structure

1.3.1 1.1 Definition

The 6-dimensional structure treated in this paper is defined as follows.

- **Geometric structure**: The combinatorial structure of the 5-dimensional hyperrectangle (vertices, edges, faces, 3-cells, 4-facets, and 5-cell body)
- **Label axis**: A discrete label axis $Q \in \{0, r, g, b\}$ with 4 values

The vertex coordinates of the 5-dimensional hyperrectangle are denoted $(x_1, x_2, x_3, x_4, x_5)$, where each coordinate **takes any real value**. When a specific proposition imposes conditions on coordinate values within the configuration structure, such conditions are declared locally within that proposition.

The 6th axis Q is not a geometric coordinate but a discrete label value associated with each state. Its range is the 4 values $\{0, r, g, b\}$. The addition of the Q axis does not change the combinatorial structure (number of vertices, edges, faces, etc.) of the hyperrectangle.

However, in the classification of bivectors that determine spin, Q participates in the enumeration of orientations as one axis on equal footing with the other 5 axes. That is, combinations such as $e_t \wedge e_Q$ and $e_R \wedge e_Q$ are abstract bivector pairs defined as axis combinations, not geometric 2-faces.

1.3.2 1.2 Element Counts of the k-Dimensional Hyperrectangle

Let $f_j(k)$ denote the number of j-dimensional faces (j-faces) of the k-dimensional hyperrectangle. We write out the element counts from low dimensions so that the reader can verify the general formula.

Vertex count $f_0(k)$: The combinatorial count distinguishing the 2 states at both ends of each axis

k	0	1	2	3	4	5
$f_0(k)$	1	2	4	8	16	32

$$f_0(k) = 2^k$$

Edge count $f_1(k)$: Extending along 1 axis, determined by the positions at both ends of the remaining $(k-1)$ axes

k	0	1	2	3	4	5
$f_1(k)$	0	1	4	12	32	80

$$f_1(k) = k \cdot 2^{k-1}$$

Verification: For $k = 3$, $3 \cdot 4 = 12$ (12 edges of a cube); for $k = 5$, $5 \cdot 16 = 80$.

Square face count $f_2(k)$: Faces spanned by 2 axes

k	0	1	2	3	4	5
$f_2(k)$	0	0	1	6	24	80

$$f_2(k) = \binom{k}{2} \cdot 2^{k-2}$$

Verification: For $k = 3$, $3 \cdot 2 = 6$ (6 faces of a cube); for $k = 5$, $10 \cdot 8 = 80$.

Cubic cell (3-cell) count $f_3(k)$: Cubic cells spanned by 3 axes

k	0	1	2	3	4	5
$f_3(k)$	0	0	0	1	8	40

$$f_3(k) = \binom{k}{3} \cdot 2^{k-3}$$

Verification: For $k = 4$, $4 \cdot 2 = 8$; for $k = 5$, $10 \cdot 4 = 40$.

4-facet count $f_4(k)$: 4-dimensional cells spanned by 4 axes

k	0	1	2	3	4	5
$f_4(k)$	0	0	0	0	1	10

$$f_4(k) = \binom{k}{4} \cdot 2^{k-4}$$

Verification: For $k = 5$, $5 \cdot 2 = 10$.

5-cell body count $f_5(k)$: For $k = 5$

$$f_5(5) = \binom{5}{5} \cdot 2^0 = 1$$

1.3.3 General Formula

Summarizing the above, the number of j-dimensional faces of the k-dimensional hyperrectangle is given by

$$f_j(k) = \binom{k}{j} \cdot 2^{k-j} \qquad (0 \leq j \leq k)$$

This corresponds to choosing j axes out of k to span the j-face ($\binom{k}{j}$ ways) and specifying which end of each of the remaining $(k - j)$ axes the face is located at (2^{k-j} ways).

Verification (Euler characteristic): $\sum_{j=0}^{k}(-1)^j f_j(k) = (2-1)^k = 1$, which is consistent with the hyperrectangle being a contractible space.

1.3.4 Values Used in This Paper ($k = 5$)

j	Element	$f_j(5)$
0	Vertices	32
1	Edges	80
2	Square faces	80
3	Cubic cells (3-cells)	40
4	4-facets	10
5	5-cell body	1

1.3.5 1.3 Axis Symmetry and Definition of Sign Configurations

1.3.1 Complete Symmetry of 5 Axes **Proposition 1.1 (Complete symmetry of 5 axes)**. The 5 geometric axes of the 5-dimensional hyperrectangle are all equivalent in their combinatorial structure. Any permutation of axes gives an automorphism of the hyperrectangle, and the 5 axes are indistinguishable with respect to the element counts $f_j(k)$ of §1.2 and subsequent geometric propositions.

Proof. The automorphism group of the combinatorial structure of the k-dimensional hyperrectangle is the hyperoctahedral group B_k, which includes all axis permutations. Therefore, the 5 axes are completely symmetric combinatorially. □

1.3.2 Axis Classification **Definition 1.2 (Axis notation labels)**. The following notation labels are assigned to the 5 axes x_1, x_2, x_3, x_4, x_5.

- x_1, x_2, x_3 are denoted x, y, z
- x_4 is denoted t
- x_5 is denoted R

By Proposition 1.1, the 5 axes are equivalent in combinatorial structure, so the assignment of labels to axes is merely a definitional choice. These labels are used as notational prerequisites for the definition of sign configurations (§1.3.3) and the definition of spin (§2).

x, y, z, t, R **are merely notational labels and do not imply geometric distinction among the axes.** The assignment of physical interpretations (space, time, curvature, etc.) to each axis is performed in §10 (an example of interpretation), which is an interpretation from outside the model and is independent of the geometric structure of §1.

1.3.3 Rigorous Definition of Sign Configurations Each component x_i $(i = 1, \ldots, 5)$ of a geometric axis is a symbol representing the **range constraint** that the coordinate value along that axis must satisfy. This paper uses the following 5 types of symbols.

Symbol	Meaning	Range of coordinate values
$+$	Positive	$x_i > 0$
$-$	Negative	$x_i < 0$
0	Origin	$x_i = 0$
$\pm$	Non-zero	$x_i \neq 0$ (positive or negative)
$*$	Arbitrary	x_i is any real number

For the Q axis, similar symbols representing the range are used.

Symbol	Meaning	Value of Q
0	Color-neutral	$Q = 0$
r, g, b	Specified color	Q is the specified color label
$r/g/b$	Any of the 3 colors	$Q \in \{r, g, b\}$
$\bar{Q}$	Anti-color	$r \leftrightarrow \bar{r}, g \leftrightarrow \bar{g}, b \leftrightarrow \bar{b}, 0 \leftrightarrow 0$
$*$	Arbitrary	$Q \in \{0, r, g, b\}$

Remark. Sign configurations are not coordinate values themselves but symbols specifying **conditions** that coordinate values must satisfy. As stated in §1.1, coordinate values in this model can take any real value, but in specific propositions and configuration structures, the above range constraints are imposed on the coordinate values of the relevant axes.

Example. The configuration $(+, +, 0, 0, 0, 0)$ refers to the set of states satisfying $x_1 > 0, x_2 > 0, x_3 = 0, x_4 = 0, x_5 = 0, Q = 0$.

1.4 2. Definition of Spin

Definition 2.1 (Spin). For a configuration (x, y, z, t, R, Q), let n denote the number of axes among t, R, Q whose sign configuration is not 0. Spin s is defined as follows.

- When 2 of the xyz axes are non-zero and 1 is 0: $s = n + 1/2$ (fermion)
- Otherwise: $s = n$ (boson)

Since t, R, Q comprise 3 axes, the possible values of n are the 4 cases $0, 1, 2, 3$. Therefore, the possible values of spin s in this model are the following 8 types.

n	Boson $s = n$	Fermion $s = n + 1/2$
0	$s = 0$	$s = 1/2$
1	$s = 1$	$s = 3/2$
2	$s = 2$	$s = 5/2$
3	$s = 3$	$s = 7/2$

Remark. This definition is a combinatorial definition internal to the model. The terms "boson" and "fermion" here are terminology within the model; correspondence to physical spin angular momentum, statistics, $(-1)^{2s}$ transformation properties, etc. is treated in §10 (an example of interpretation).

1.5 3. Enumeration of Q-Label Transitions

Consider transitions among the 3 non-zero values $\{r, g, b\}$ of the Q axis. A transition is described by the pair $(Q_{\text{initial}}, Q_{\text{final}})$, and the total number of transitions from 3 values to 3 values is

$$3 \times 3 = 9$$

Among these, one linear combination of the diagonal components (the state proportional to $r\bar{r} + g\bar{g} + b\bar{b}$) has the same action on all color labels and does not mediate label transitions. Excluding this 1 state,

$$3 \times 3 - 1 = 8$$

is the number of independent transition states. Physical naming is given in §10.

1.6 4. Enumeration of Spin-1 Configurations

Spin-1 configurations have $n = 1$ by Definition 2.1, meaning exactly 1 of the t, R, Q axes has a non-zero sign configuration, while all xyz axes are 0.

Depending on which of t, R, Q is non-zero, the number of configuration states is classified as follows.

Non-zero axis	Configuration form	Number of states
t	$(0, 0, 0, \pm, 0, 0)$	2 (2 choices of sign $\pm$)
R	$(0, 0, 0, 0, \pm, 0)$	2 (2 choices of sign $\pm$)
Q	$(0, 0, 0, 0, 0, Q_{\text{initial}} \to Q_{\text{final}})$	9 (by §3)

Total: $2 + 2 + 9 = 13$ states (for the interpretation excluding 1 state, see §10)

1.7 5. Enumeration of Spin-2 Configurations

Spin-2 configurations have $n = 2$ by Definition 2.1, meaning exactly 2 of the t, R, Q axes have non-zero sign configurations, while all xyz axes are 0.

The number of ways to choose 2 axes from the 3 axes t, R, Q is $\binom{3}{2} = 3$, and the configurations are classified according to the combination of 2 non-zero axes as follows.

Non-zero 2 axes	Configuration form
t, R	$(0, 0, 0, \pm, \pm, 0)$
t, Q	$(0, 0, 0, \pm, 0, Q)$
R, Q	$(0, 0, 0, 0, \pm, Q)$

Therefore, the number of types of spin-2 configurations in this model is **3**.

1.8 6. Enumeration of Spin-3 Configurations

Spin-3 configurations have $n = 3$ by Definition 2.1, meaning all of t, R, Q are non-zero, while all xyz axes are 0.

The number of combinations where all 3 axes t, R, Q are non-zero is $\binom{3}{3} = 1$.

Non-zero 3 axes	Configuration form
t, R, Q	$(0, 0, 0, \pm, \pm, Q)$ $(Q \neq 0)$

Therefore, the number of types of spin-3 configurations in this model is **1**.

1.9 7. Sign Product of Non-Zero Axes

Definition 7.1 (Sign product). For a boson configuration, when the n axes among t, R, Q with non-zero sign configurations each take the sign $+$ or $-$, the product of these signs is defined as the **sign product** $P(n)$.

$$P(n) = (-1)^n$$

n	$P(n)$
0	$+1$
1	-1
2	$+1$
3	-1

Proposition 7.1. $P(n)$ depends only on the parity of n. When n is even, $P(n) = +1$; when n is odd, $P(n) = -1$.

Remark. $n = 0$ (no sign reversal) and $n = 2$ (double reversal) give the same $P = +1$, but their algebraic structures differ. $n = 0$ is the identity transformation, while $n = 2$ is the self-inverse of sign reversal.

The physical interpretation of $P(n)$ (correspondence to force directionality) is treated in §10 (an example of interpretation).

1.10 8. Signed Area of Square Faces Containing the R Axis

1.10.1 8.1 Definition of Signed Area

Definition 8.1 (Signed area). The **signed area** $M(\sigma)$ of a configuration structure σ is defined as the sum of the products of sign configurations and areas over all square faces containing the R axis.

$$M(\sigma) = \sum_{R\text{-containing square face } f} \operatorname{sgn}(f) \cdot \operatorname{Area}(f)$$

$M(\sigma)$ is a fundamental conserved quantity of this model. Physical interpretation (such as analogy with mass) is treated in §10.

1.10.2 8.2 Classification of Faces Containing the R Axis

Among the 80 square faces, the faces containing R ($= x_5$) are classified as follows.

Face type	Bivector	Number of types	Number of faces
Spatial$-R$	$e_i \wedge e_5$	3	$3 \times 8 = 24$
$t - R$	$e_4 \wedge e_5$	1	8
Total		4	**32**

The faces containing the R axis are 32 out of 80 total faces, constituting 2/5 of the total.

1.10.3 8.3 Existence of Configurations with $M(\sigma) = 0$

Proposition 8.1. Among the configuration structures σ, there exist those for which $M(\sigma) = 0$ (the signed area completely cancels).

The interpretation of configurations with $M(\sigma) = 0$ and their physical correspondence is treated in §10.

1.11 9. Conclusions of the Main Body

In this paper, we defined a 6-dimensional structure obtained by adding a discrete label axis Q to the 5-dimensional hyperrectangle, and derived the following combinatorial properties from its configuration structure.

1. **Element counts of hyperrectangles** (§1.2): The number of j-dimensional faces of the k-dimensional hyperrectangle is $f_j(k) = \binom{k}{j} \cdot 2^{k-j}$. For $k = 5$: 32 vertices, 80 edges, 80 square faces, 40 cubic cells, 10 4-facets.
2. **Complete symmetry of 5 axes** (Proposition 1.1): The 5 geometric axes are all equivalent in combinatorial structure, and any axis permutation gives an automorphism of the hyperrectangle.
3. **Axis notation labels** (Definition 1.2): The notation labels x, y, z, t, R are assigned to the 5 axes.
4. **Rigorous definition of sign configurations** (§1.3.3): Range constraints on each axis are expressed using the 5 symbols $+, -, 0, \pm, *$.
5. **Definition of spin** (Definition 2.1): Spin s is classified into 8 types ($s = 0, 1/2, 1, 3/2, 2, 5/2, 3, 7/2$) from the configuration (x, y, z, t, R, Q), based on the number n ($= 0, 1, 2, 3$) of non-zero axes among t, R, Q and the structure of the xyz axes.
6. **Enumeration of Q-label transitions** (§3): Transitions from 3 colors to 3 colors give $3 \times 3 = 9$.
7. **Enumeration of spin-1 configurations** (§4): 2 states with t fixed, 2 states with R fixed, 9 states with Q fixed, totaling 13 states.
8. **3 types of spin-2 configurations** (§5): tR, tQ, RQ — 3 types.
9. **1 type of spin-3 configurations** (§6): tRQ — 1 type.
10. **Sign product** $P(n) = (-1)^n$ (Definition 7.1, Proposition 7.1): The product of signs of non-zero axes depends only on the parity of n.
11. **Signed area** $M(\sigma)$ (Definition 8.1): A fundamental conserved quantity of the model, defined from the 32 square faces containing the R axis. Configurations with $M(\sigma) = 0$ exist.

All of these are results derived purely from combinatorics and geometry under the external assumptions of this model (adoption of the 5-dimensional hyperrectangle, addition of the Q axis).

1.12 10. An Example of Interpretation: Correspondence with the Standard Model

The content of this section is not a claim of this paper but an example of interpretation. It does not attempt to physically derive the Standard Model from this model. This section presents an example of how the 6-dimensional structure and its mathematical properties defined and derived in §1–§9 might be mapped to the particle classification of the Standard Model.

The following correspondences are not justified from within the model and are **choices of interpretation from outside**. If an alternative mapping becomes available in the future, it can be added in parallel with this section.

1.12.1 10.1 Assignment of Physical Labels

As stated in §1.3.2, the 5 axes are all equivalent in combinatorial structure, but in the mapping to the Standard Model, the following labels are assigned.

- $x_1, x_2, x_3 \to xyz$: 3-dimensional spatial axes
- $x_4 \to t$: Time axis
- $x_5 \to R$: Curvature axis (corresponding to the curvature radius of space)
- Q: Color label axis (0 = color-neutral, r, g, b = color charge)

1.12.2 10.2 Correspondence to Quarks/Leptons

The classification of particles corresponds to the value of the Q axis as follows.

Q value	Correspondence
0	Lepton (no color charge)
r, g, b	Quark (color charge present)

For the same geometric configuration (x, y, z, t, R), the existence of 4 possible values of Q causes each generation of fermions to branch into 1 lepton type and 3 quark colors.

Charge correspondence: The sign of electric charge is distinguished by the sign configuration of the spatial axes. The denominator 3 of the fractional quark charges $(\pm 1/3, \pm 2/3)$ coincides with the number of non-zero values of the Q axis ($|\{r, g, b\}| = 3$). The complete derivation of charge values, particularly the asymmetry between $+2/3$ and $-1/3$, remains an open problem (§10.10).

Consideration of a 7th axis: The design of adding electric charge as an independent 7th axis was also considered, but increasing the number of non-spatial axes to 4 would increase the number of $n = 2$ types to $\binom{4}{2} = 6$, producing a particle spectrum similar to supersymmetric theories. Given the current experimental situation (non-discovery of supersymmetric particles), this paper adopts the approach of reading electric charge from the structure of the Q axis and the signs of the spatial axes.

1.12.3 10.3 Correspondence to 8 Gluon States

Of the 9 states of Q-label transitions enumerated in §3, the **8 states excluding the color singlet ($r\bar{r} + g\bar{g} + b\bar{b}$)** are mapped to **gluons** $g_1, \ldots, g_8$.

In the Standard Model, the 8 gluon states are derived as the 8-dimensional adjoint representation of the Lie algebra $\mathfrak{su}(3)$ of SU(3). In this interpretation, the same number 8 arises

combinatorially from the 4-label structure of the Q axis, but the non-commutative structure constants f^{abc} of SU(3) are not derived from this model.

1.12.4 10.4 Correspondence to 12 Gauge Boson States

Of the 13 spin-1 configuration states enumerated in §4, the 8 states obtained by excluding the color singlet from the 9 Q-fixed states, together with the 2 t-fixed states and 2 R-fixed states, give **12 states mapped to gauge bosons**.

Fixed axis	Correspondence	Number of states
t	W^+, W^-	2
R	γ, Z^0	2
Q	$g_1, \ldots, g_8$	8
Total		**12**

The 2 states of the W boson ($t = +1, -1$) correspond to the positive and negative charges, while γ and Z^0 correspond to the presence or absence of $M(\sigma)$ due to the sign of the R axis ($M(\gamma) = 0, M(Z) \neq 0$). The total of 12 matches the Standard Model gauge boson state count $\gamma + g \times 8 + W^+ + W^- + Z^0 = 12$.

1.12.5 10.5 Interpretation of Spin-2 Configurations

The following correspondences are given to the 3 types of spin-2 configurations enumerated in §5.

Type	Fixed axes	Correspondence
tR type	t, R	**Graviton** (correspondence within the Standard Model + gravity framework)
tQ type	t, Q	No known particle in the Standard Model
RQ type	R, Q	No known particle in the Standard Model

By Proposition 7.1, all 3 types have $P(2) = +1$, and under the interpretation of §10.7, they mediate only attractive forces. For the tR type, this is consistent with the property of gravity (attraction only). The tQ and RQ types are states whose existence is derived from the combinatorial structure of this model, but have no counterpart in the current Standard Model.

1.12.6 10.6 Interpretation of Spin-3 Configurations

The 1 type (tRQ) of spin-3 configurations enumerated in §6 has no known particle counterpart in the Standard Model.

1.12.7 10.7 Correspondence Between the Sign Product $P(n)$ and Force Directionality

The following physical interpretation is given to the sign product $P(n) = (-1)^n$ defined in §7.

- $P(n) = +1$ (n even): Force is unidirectional (**attraction only**)
- $P(n) = -1$ (n odd): Force is bidirectional (**both attraction and repulsion**)

n	$P(n)$	Interpretation of force directionality
0	$+1$	No direction (scalar field, $s = 0$)
1	-1	Both attraction and repulsion ($s = 1$)
2	$+1$	Attraction only ($s = 2$)
3	-1	Both attraction and repulsion ($s = 3$)

Under this interpretation, all 3 types of spin-2 configurations (tR, tQ, RQ) have $n = 2$ (even) and mediate only attractive forces. This is consistent with the property of gravity (attraction only). It is also consistent with the electromagnetic force at spin 1 ($n = 1$, both attraction and repulsion).

Remark. This correspondence is not justified from within the model, and the physical mechanism connecting the parity of $P(n)$ to force directionality is not derived in this paper.

1.12.8 10.8 Range of the R Axis and Force Coupling Strengths

Range of the R axis: In prior work [W5], the number-theoretic conditions of central projection in discrete space lead to the discretization $R = \pm n^2 \cdot L_0$ ($n \in \mathbb{N}$, L_0 is the unit length). However, the results of this paper do not require this discretization and hold even when R takes continuous values. Interpreting the range of the R axis as unbounded corresponds to the unboundedness of the mass spectrum.

Interpretation of each force and the axis correspondence in this model:

Force	Mediating particle	Involved axis	Coupling characteristics
Strong force	g_1–g_8	Q axis (8 color transition channels)	Between color-charged particles only
Electromagnetic force	γ	R axis	Between non-zero charged particles
Weak force	$W^{\pm}, Z^0$	t axis	All fermions
Gravity	G	$t \wedge R$ ($n = 2$)	All particles (Q-independent)

Remark. The coupling constants in the Standard Model near M_Z are $\alpha_s \approx 0.1$, $\alpha \approx 1/137$, $\alpha_W \approx 1/30$, $\alpha_G \approx 10^{-39}$; the strong, electromagnetic, and weak forces are of similar order, while gravity alone is extremely weak. The quantitative derivation of this hierarchy is an open problem for future work.

1.12.9 10.9 Analogy Between $M(\sigma)$ and Mass

The signed area $M(\sigma)$ defined in §8.1 has the following **analogies** with particle mass.

- Configurations with $M(\sigma) = 0$ can be interpreted as corresponding to massless particles (photon, gluons)
- The unboundedness of the R axis range corresponds to the unboundedness of the mass spectrum

However, this paper does not identify $M(\sigma)$ with mass. The derivation of specific mass values is an open problem for future work (§10.12).

1.12.10 10.10 Table A: Correspondence Between Confirmed Particles and 6-Dimensional Configurations

Bosons:

Particle	Symbol	s	Configuration (x, y, z, t, R, Q)	n	Mass (MeV)
Higgs	H^0	0	$(\pm, 0, 0, 0, 0, 0)$ etc.	0	125,250
W boson	W^+	1	$(0, 0, 0, +, 0, 0)$	1	80,379
W boson	W^-	1	$(0, 0, 0, -, 0, 0)$	1	80,379
Photon	γ	1	$(0, 0, 0, 0, +, 0)$	1	0
Z boson	Z^0	1	$(0, 0, 0, 0, -, 0)$	1	91,188
Gluon	g_1–g_8	1	$(0, 0, 0, 0, 0, Q_i \to Q_j)$	1	0
Graviton	G	2	$(0, 0, 0, \pm, \pm, 0)$	2	0

Leptons ($Q = 0$):

Particle	Symbol	s	Generation	Configuration	Mass (MeV)
Electron	e^-	1/2	1	$(+, +, 0, 0, 0, 0)$	0.511
Electron neutrino	ν_e	1/2	1	$(+, -, 0, 0, 0, 0)$	$< 10^{-6}$
Muon	μ^-	1/2	2	$(+, 0, +, 0, 0, 0)$	105.66
Muon neutrino	ν_μ	1/2	2	$(+, 0, -, 0, 0, 0)$	< 0.17
Tau	τ^-	1/2	3	$(0, +, +, 0, 0, 0)$	1,776.9
Tau neutrino	ν_τ	1/2	3	$(0, +, -, 0, 0, 0)$	< 15.5

Anti-leptons correspond to configurations with the signs of the spatial axes reversed.

Quarks ($Q \in \{r, g, b\}$):

Particle	Symbol	s	Generation	Configuration	Mass (MeV)
Up	u	1/2	1	$(+, +, 0, 0, 0, r/g/b)$	2.16
Down	d	1/2	1	$(+, -, 0, 0, 0, r/g/b)$	4.67
Charm	c	1/2	2	$(+, 0, +, 0, 0, r/g/b)$	1,270
Strange	s	1/2	2	$(+, 0, -, 0, 0, r/g/b)$	93.4
Top	t	1/2	3	$(0, +, +, 0, 0, r/g/b)$	172,760
Bottom	b	1/2	3	$(0, +, -, 0, 0, r/g/b)$	4,180

Each quark has 3 color states $Q = r, g, b$. Anti-quarks correspond to sign-reversed spatial axes and anti-colors of Q.

Summary (states with $s = 0, 1/2, 1, 2$ mappable to the SM):

Classification	s	Q	Model-derived count	SM correspondence
Higgs	0	0	1	1
Gauge boson (t type)	1	0	2	W^+, W^-

Classification	s	Q	Model-derived count	SM correspondence
Gauge boson (R type)	1	0	2	γ, Z^0
Gluon (Q type)	1	transition	9	8 (excluding 1 color singlet)
Graviton (tR type)	2	0	1	G (beyond SM)
Lepton	1/2	0	6	6
Anti-lepton	1/2	0	6	6
Quark	1/2	r, g, b	18	18
Anti-quark	1/2	$\bar{r}, \bar{g}, \bar{b}$	18	18
Subtotal			**63**	**62 (61 + G)**

(Model: 63 states = 62 SM-mapped states + 1 color singlet state). Of the 63 states, 62 are mapped to the Standard Model's 61 states + 1 graviton state. The remaining 1 state (the color singlet of the Q transition) is derived by the model but does not correspond to an independent gauge boson in the Standard Model.

In addition, configurations with $s = 3/2, 2, 5/2, 3, 7/2$ (Table B, §10.11) are derived from the model, but no known particles in the Standard Model correspond to them.

1.12.11 10.11 Table B: Configurations Without Standard Model Counterparts

The following lists states that are derived from the combinatorial structure of this model but have no known particle counterpart in the Standard Model. These are combinatorial consequences within the model and do not assert the existence of undiscovered particles.

Bosons (xyz all 0):

Type	s	n	Configuration form	Number of states
tQ type	2	2	$(0, 0, 0, \pm, 0, Q_i \to Q_j)$	$2 \times 9 = 18$
RQ type	2	2	$(0, 0, 0, 0, \pm, Q_i \to Q_j)$	$2 \times 9 = 18$
tRQ type	3	3	$(0, 0, 0, \pm, \pm, Q_i \to Q_j)$	$4 \times 9 = 36$

Fermions (2 of xyz non-zero, 1 is 0):

Type	s	n	Non-zero non-spatial axes	Spatial-side pattern count	Q values
t type	3/2	1	t	$3 \times 4 = 12$	4
R type	3/2	1	R	12	4
Q type	3/2	1	Q (non-zero)	12	3
tR type	5/2	2	t, R	12	4
tQ type	5/2	2	t, Q	12	3
RQ type	5/2	2	R, Q	12	3
tRQ type	7/2	3	t, R, Q	12	3

Remark. The spatial-side pattern count of 12 arises from $\binom{3}{1} = 3$ ways to choose the zero axis $\times\ 2^2 = 4$ sign choices for the 2 non-zero axes. The Q values are 4 when $Q = 0$ is included, and 3 ($\{r, g, b\}$) when the Q axis is required to be non-zero.

1.12.12 10.12 Open Problems in This Interpretation

The following are open problems within the scope of this section's interpretation.

10.12.1 Quantitative Computation of $M(\sigma)$ and Mass Ratios

Explicitly computing the signed area $M(\sigma)$ for each configuration structure and comparing with known particle mass ratios is a means of testing this interpretation.

- $m_\mu/m_e \approx 206.8$: Mass ratio between generations
- $m_W/m_Z \approx 0.881$: Relation to the Weinberg angle
- $m_t/m_e \approx 3.4 \times 10^5$: Maximum mass ratio

10.12.2 Derivation of Gauge Groups

A correspondence is suggested between the 4-label structure of the Q axis and SU(3), the 2 values of the t axis and SU(2), and the continuous scale of the R axis and U(1), but the rigorous mechanism for the transition from discrete structure to continuous gauge groups is unproven. In particular, the number of gluons 8 is reproduced, but the information of the non-commutative structure constants f^{abc} is not derived.

10.12.3 CKM Matrix and PMNS Matrix

How inter-generational mixing (CKM matrix, PMNS matrix) is derived as a breaking of S_3 symmetry remains unsolved.

10.12.4 Higgs Mechanism

The Higgs boson corresponds to $n = 0$ (spatial axes only), but how spontaneous symmetry breaking by the Higgs field is understood within this framework is an open problem for future work.

10.12.5 Complete Derivation of Charge Values

The sign of electric charge can be read from the sign configurations of the spatial axes, and the unit of the absolute value of charge (1 vs 1/3) from the Q axis sector, but the asymmetry between $+2/3$ and $-1/3$ is the most prominent open problem of this interpretation.

10.12.6 Additional Predictions of $s = 2, 3$ Bosons

If tQ type and RQ type spin-2 bosons, and the tRQ type spin-3 boson exist, their experimental consequences need to be clarified.

1.12.13 10.13 Conclusion of the Interpretation Example

This section presented an example mapping the mathematical properties of the 6-dimensional structure defined and derived in §1–§9 to the particle classification of the Standard Model.

1. Of the 8 spin types ($s = 0, 1/2, 1, 3/2, 2, 5/2, 3, 7/2$), 62 of the 63 states with $s = 0, 1/2, 1, 2$ can be mapped to known particles of the Standard Model (61 + graviton) (the remaining 1 state is the color singlet)
2. The 4 values of the Q axis correspond to color charge and the lepton/quark branching
3. Of the 9 Q-label transitions, 8 states correspond to gluons (the color singlet has no counterpart)
4. Of the 13 spin-1 configuration states, 12 correspond to Standard Model gauge bosons
5. The tR type of spin-2 configurations corresponds to the graviton. The tQ and RQ types are derived from the model but have no Standard Model counterpart
6. The tRQ type of spin-3 configurations similarly has no counterpart
7. Fermion types with $s = 3/2, 5/2, 7/2$ also exist combinatorially but do not correspond to known particles (Table B)
8. $M(\sigma)$ has analogies with particle mass
9. The types of axes involved in each force can be interpreted as the qualitative origin of the coupling strength hierarchy

This model derives configurations that **exceed** the known particles of the Standard Model. These "surplus states" are open problems in this interpretation and will be examined in future research as one of the following: (a) predictions of undiscovered particles, (b) combinations not physically realized (existence of selection rules), or (c) limitations of this interpretation.

The derivation of specific mass values, charge values, non-commutative structure of gauge groups, and generation mixing matrices remains an open problem for future work (§10.12).

1.13 References

[1] Finkelstein, D. & Rubinstein, J. "Connection between Spin, Statistics, and Kinks." *J. Math. Phys.* 9, 1762 (1968).

[2] Atiyah, M.F., Manton, N.S. & Schroers, B.J. "Geometric Models of Matter." *Proc. R. Soc. A* 468, 1252 (2012). arXiv: 1108.5151.

[3] Furey, C. "Standard Model Physics from an Algebra?" arXiv: 1611.09182 (2016).

[4] Kaluza, Th. "Zum Unitätsproblem der Physik." *Sitzungsber. Preuss. Akad. Wiss. Berlin* 966–972 (1921).

[5] Coxeter, H.S.M. *Regular Polytopes.* Dover (1973).

[6] Kihara, N. "Bivectorial Classification of Spin Derived from the Orientation Structure of the 5-Dimensional Hypercube." DOI: 10.5281/zenodo.19643358 (2026).

[7] Kihara, N. "Geometric Classification of Spin Derived from the Orientation Structure of the 5-Dimensional Orthoplex." DOI: 10.5281/zenodo.19630972 (2026).

[W5] Kihara, N. "Complete Spherical Coverage of Central Projection in Discrete Space and the Number-Theoretic Necessity of the 5-Dimensional Background Space." DOI: 10.5281/zenodo.19624957 (2026).